

1. (b) Data and a pointer to the next node.
2. (c) NULL
3. (b) Self-referential structure.
4. (b) head == NULL
5. (b) newnode $\rightarrow$ next = head; head = newnode;
6. (b) temp = head; head = head $\rightarrow$ next; free(temp);
7. (c) Traversing from tail to head.
8. (b) n.
9. (c) Copying the data of p $\rightarrow$ next into p, then deleting p $\rightarrow$ next.
10. (b) While (P != NULL)
11. (b) P $\rightarrow$ next == NULL
12. (c) Two pointers advancing at different rates.
13. (c) Three
14. (b) Exactly one node.
15. (b) P = (struct node*) malloc (sizeof (struct node));
16. (b) Avoid special-case handling for insertion and deletion at the head.
17. (c) The first node.
18. (b) It does not support random (indexed) access.
19. (b) Loss of access to all nodes, causing a memory leak.
20. (b) A slow pointer and a fast pointer.
21. (c) The last node.
22. (b) Nodes may be scattered anywhere in memory.

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23.

```
struct node {  
    int data;  
    struct node *next;  
};
```

24.

Given list:

10 → 20 → 30 → 40 → NULL

The program adds each node with the next node.

$$10 + 20 = 30$$

$$20 + 30 = 50$$

$$30 + 40 = 70$$

Output: 30 50 70

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~~25. void insertEnd(struct node *head, int x) {
 struct node *t~~

25. if (*head == NULL) {
 *head = t;
 return;
}

26. struct node * reverse(struct node *head) {
 struct node * prev = NULL;
 struct node * curr = head;
 struct node * next;

while (curr != NULL) {
 next = curr → next;
 curr → next = prev;
 prev = curr;
 curr = next;
}
return prev;
}

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27. Suppose we have
 $10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$

To delete 20:

$\text{prev} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$
 $\text{free}(\text{temp});$

After deletion,

$10 \rightarrow 30 \rightarrow \text{NULL}.$

28. Initial list:

$5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$

(i) insert 7 at beginning

$7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$

(ii) Delete 12

$7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow \text{NULL}$

(iii) insert 9

$7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$

final list

$7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$